

AUDRIC2: A Modular and Highly Interconnected Automated Driving Framework Focus on Decision Making and Vehicle Control

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Abstract—Last years, automated vehicle technologies made considerable progress in academia and the industry. However, there is still a need for more scalable solutions which ensure fast progress between the prototypes and the final deployments. In this sense, this work introduces the updated version of the AUtomed DRIVING Core framework (AUDRIC), which aims for robust and reliable vehicle decision-making and control algorithms. The solution is modular and provides MATLAB and ROS interfaces. Also, it has the integration links between the CARLA simulator and Dynacar (in-house simulator). The Autoware.AI framework was integrated to support vehicle perception. This framework targets non-holonomic ground vehicles, such as passenger cars, buses, shuttles, non-holonomic industrial AGV, etc. A simulation around TecNALIA premises was used to verify the provided functionalities (Basque Country, Spain). This framework will continue its development with the support of the European project SHOW. Vehicle cooperation functions are considered as an upcoming feature.

I. INTRODUCTION

Automated vehicles have shown great potential for improving safety, vehicle comfort, and efficiency during driving tasks. Nevertheless, robust and optimal solutions for vehicle decision and control are demanded in compliance with the geometry of the maps, vehicle dimensions, its dynamics, and safety [1].

The decision making processes of the vehicle has been described, classically, as a hierarchical structure where some of the layers are: i) routing based on requirements of the passengers, traffic conditions, and road network; ii) reasoning through a sequence of driving behaviors such as turning, lane-changing, among others; and iii) the generation of feasible trajectories that lead the vehicle safely [2]. Additionally, the control layer is in charge of tracking the navigation commands provided by vehicle planning through the trajectory (path and speed profile) [3].

Since the DARPA challenges and even more recently, many groups and companies have tried to standardized frameworks for automated driving obtaining good results. Moreover, newer approaches are looking for more general and scalable solutions that bring prototypes to commercial deployments.

In these terms, a simulation framework for testing and evaluating cooperative intelligent transportation system (C-ITS) was proposed [4]. The approach combines the driv-

ing simulation software from the Swedish National Road and Transport Research Institute (VTI) and a framework for traffic and network simulation specifically designed for testing platoon applications (Platooning Extension for Veins, or Plexe). The strength of the approach relies on combining three different simulators providing the best advantages of each one during testing.

Another approach proposed a framework for sensor fusion of self-driving cars and other autonomous robots [5]. The main goal of the contribution was to design a universal and scalable platform for building up robust 3D models of the agent's surrounding environment by fusing a wide range of various sensors. The authors have considered RGB and thermal cameras, 3D LiDARs, 3D IMU, and a GNSS. The framework covers a complete pipeline from data loading, filtering, pre-processing, environment model construction, visualization, and data storage. The software was design to be compatible with ROS. Nevertheless, high performance, high concurrency, and multi-threading were not considered because it is looking for a clear architecture, easy to scale up, and simple pipeline.

Down in the control hierarchy, when it comes to decision-making based on the data gotten from the perception system, there are several recent proposals. In [6] probabilistic methods are used to evaluate the possibility of collision to assess and make safe decisions on highway situations. In their implementation with ROS, they have access to information about the pilot state (e.g., whether they are distracted or healthy). The driver state permits to implement this in the risk calculus and taking action in consequence.

Some other approaches have achieved partially scalable and replicable solutions in multiple platforms. On the project SmartShuttle, an automated driving architecture has unified software, hardware, control, and decision targeting low-speed shuttles [7]. This approach was validated in a Ford Fusion and a Dash EV. Later on, the approach was tested in another two vehicles of the same model (newer versions).

A newer approach has presented an architecture for automated cars based on ROS2 [8]. The main goal has been to achieve the demands of real-time, high reliability, and requisites of functional safety, along with ROS benefits such as standardized messages and distributed architectures.

Considering the literature, this work presents the updated version of the AUDRIC (AUtomed DRIVING Core) framework previously introduced in [9]. This version comes with a modular scheme designed to work with different real and virtual platforms. Moreover, the first developed modules are described which guarantee safe guidance of the vehicle under

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mixed traffic circumstances, using the ROS or MATLAB interfaces provided in the framework. A simulation test case, around Tecnalia premises (Basque Country, Spain), was proposed to demonstrate the benefits of the proposal. The continuous development and integration of the framework of the supported interfaces (ROS and Matlab) are the main differences compared to the literature. The European project SHOW will be used to finance -partially- the development of the ROS and Matlab interfaces. In the future, an interface for ROS2 will be included (subject to interest and budget).

The rest of the work is structured as follows, section II presents the explanation of the second version of the AUDRIC framework, section II gives details about each module and the interfaces and integration links with simulation environments and other frameworks. In section III, several validation scenarios, in the framework of the European project SHOW, are explained and the results of the first simulation tests are presented. Finally, the summary of conclusions future works is presented in the section IV.

II. AUDRIC: AUTOMATED DRIVING CORE V2.0

The AUDRIC framework was presented for the first time in 2017 [9]. However, it received its name some months later. It used to contain basic methods (e.g. static trajectories, simple controllers, etc) to guide the vehicle over a predefined route which was the basis for developing a long number of European and local projects [10]. Nevertheless, great changes have arisen in the automated driving field, as we mentioned in the introduction. This demands an improvement of AUDRIC to meet new minimum needs of the technology, such as:

- Modular tasks.
- Safe guidance of the vehicle under mixed traffic circumstances (it could not consider overtaking necessarily).
- Comfort and smoothness during a normal operation.
- Capacities to detect and modelling the environment, as well as communication with infrastructure and other vehicles on the road.
- The use of complex simulation environments, during the testing stage, that consider more details on the road, e.g. buildings, pedestrians' movement, other vehicles, etc.

Moreover, we have included another requirement which is standardizing the solution for all ground vehicles with non-holonomic constraints, at the moment, without considering a load connected with a joint, e.g. trailers, trucks, etc.

Some of the simulation environments and vehicles tested with the AUDRIC2 framework, or some of its modules, are: Dynacar simulation environment (top-left Fig. 1), CARLA simulator (top-right Fig. 1), a modified Renault Twizy (middle-left Fig. 1), a double Ackermann ground vehicle (middle-right Fig. 1), and it is expected to be tested in upcoming month on an electric Gulliver shuttle (bottom-left Fig. 1), and a 12m-long electric bus (bottom-right Fig. 1). However, this topic will be covered in more detail on section III.

In these terms, AUDRIC2 has been developed as a C++ library preserving its initial six-modules architecture including

[9]:

- Acquisition: collecting sensor information.
- Perception: vehicle and environment modeling using sensor information and communications.
- Communication: interaction among infrastructure, pedestrians, and vehicles (V2X).
- Decision: algorithms for high-level decision making, trajectory generation, collaboration among vehicles, etc.
- Control: vehicle tracking controllers. They can be coupled (steering and pedals in one controller) or decoupled (steering and pedals in different controllers).
- Actuation: interaction of the control signals with the vehicle interface to move the actuators, e.g. electric signals to external motors coupled to the pedals or steering wheel, CAN-Bus messages to interact with X-by-wire components, etc.

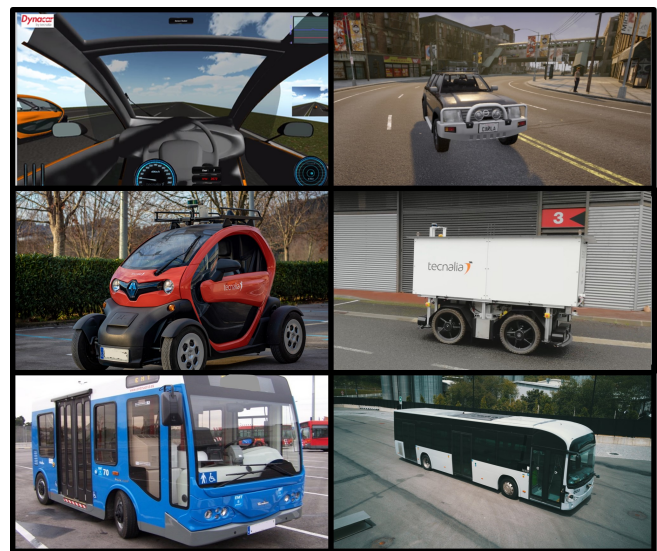


Fig. 1. Platforms using AUDRIC. (top-left). (top-right) ². (middle-left). (middle-right). (bottom-left). (bottom-right).

The acquisition and actuation modules rely on the hardware/software architecture of the considered platform. The state of the art has impressive approaches to generate vehicle positioning and inertial values, as well as environment perception; due to these reasons, the perception module is integrated from an external source and is bridged to AUDRIC. Communication is solved by standardized devices (V2X). However, the bridge between the AUDRIC information and the V2X devices is considered as part of the framework as well as internet-based data exchanging, e.g. exchange of MQTT messages with an external Data Management Platform. Considering these conditions, the framework is focused, mainly, on decision-making and vehicle control solutions which are easier to escalate and to generate a modular framework that targets non-holonomic automated vehicles.

²Image extracted from https://carla.readthedocs.io/en/0.9.11/adv_rendering_options/

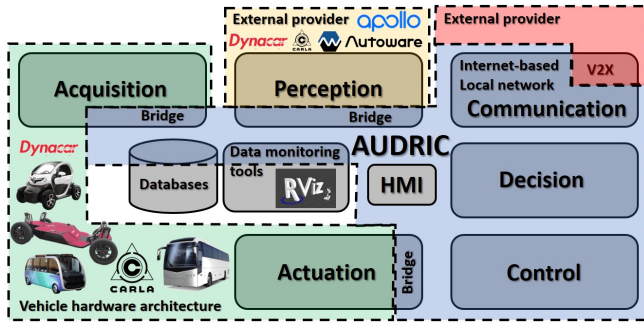


Fig. 2. Categories (modules) into the AUDRIC Framework

An abstraction of the framework, divided into stages or modules, is presented in Fig. 2. The currently available modules will be detailed in the following sections as well as the interfaces used to deploy the framework.

A. Modules

1) *Global Road Planner*: This function is part of the decision module (Fig. 2) and it plans the global route that leads the vehicle to its destination. The module relies on map information provided via an OSM file which uses a format inspired by the Lanelet2 standard [11], nevertheless, it is a simplified version of the aforementioned.

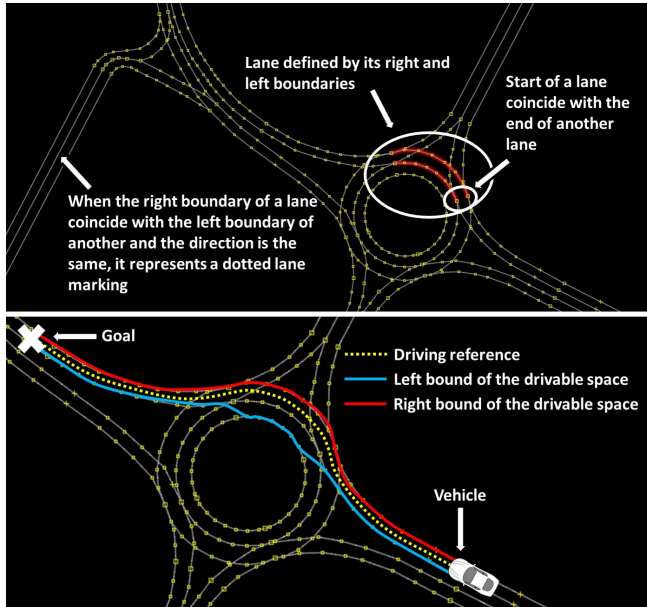


Fig. 3. Road definition

The top of Fig. 3 depicts an area of a map described by lanes. These lanes are defined as a relationship composed of a right and a left boundary where those are defined by polylines (array of points). A continuous road, which has been made from lane pieces, must have the starting nodes (boundaries) of the current lane equal to the ending nodes of the previous lane. Moreover, the relationship contains the speed limit and the presence of a yield, stop sign, or pedestrian crossing at the end of the lane.

The module is in charge of guiding the vehicle with the generation of a drivable space (bottom of Fig. 3) which has the outer right and left boundaries of the road, and the reference of the center of the lane that is more in the right side of the road. This ensures safe driving without compromising the traffic flow. This reference can be changed to the center of another lane, based on some criteria that will be studied in the future, e.g. merging, platoon, etc.

Moreover, the drivable space is obtained considering the shortest path from the vehicle to the goal considering the connections of the road map.

2) *Trajectory Generator*: This function is part of the decision module (Fig. 2) and it uses the information of the drivable space as well as the lane reference and speed reference from the *globalRoadFollower* to generate safe, smooth, and continuous trajectories (path and speed) to be tracked by the vehicle controllers.

The module uses motion primitives derived from Bézier curves based on previous approaches presented by the group [12] [13].

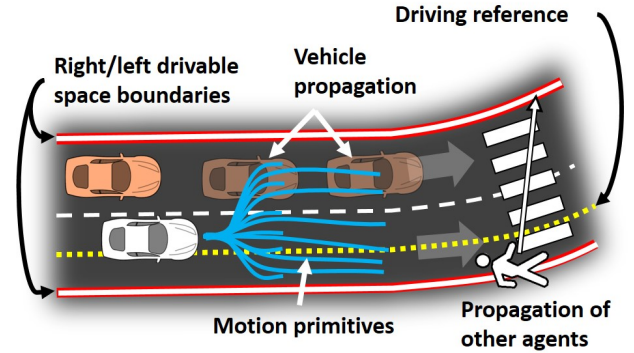


Fig. 4. Trajectory generation

This method generates layers of motion primitives as it is presented in Fig. 4. The best trajectory candidate is given by weighting the farther it reaches, the smoothness of the trajectory, and the closeness to the driving reference. Moreover, the computation of the travel distance considers, in the case of vehicles, the propagation of other agents on the road. In the case of agents out of the road limits, the propagation is performed following the same direction of the agent. Additionally, the trajectory will cruise the vehicle, totally (considering the vehicle geometry), into the drivable space boundaries.

A separate publication is intended to explain all the details of the approach as well as performing extensive tests considering specific and real driving scenarios.

3) *Control Lateral Model Based*: This solution is part of the control module (Fig. 2) and it is a lateral tracking controller (AUDRIC provides decoupled lateral and longitudinal motion controllers). It is inspired by the method of Model Predictive Control, nevertheless, the solution space is limited to a set of possible motion candidates based on the rate of change of the steering angle ($C_1, \dots, C_i, \dots, C_n$ on Fig. 5).

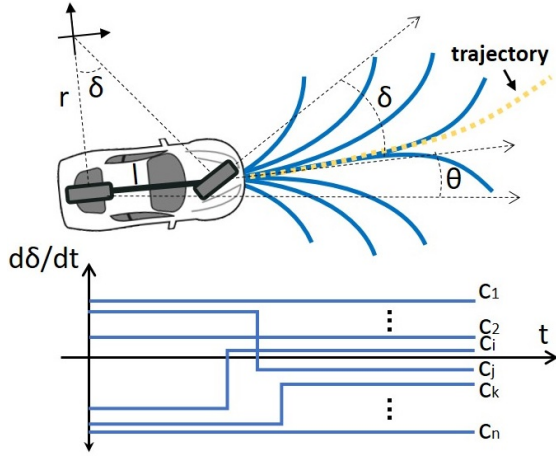


Fig. 5. Lateral controller based on steering wheel candidates

The model used is a kinematic bicycle model, as it is shown in Fig 5. The set of functions defining the model are the following:

$$\dot{\theta} = \omega = \frac{v}{r} \quad \& \quad \tan(\delta) = \frac{l}{r} \quad \rightarrow \quad \dot{\theta} = \frac{v \tan(\delta)}{l} \quad (1)$$

$$\theta = \theta_0 + \omega \Delta t \quad (2)$$

$$x = x_0 + v \Delta t \cos(\theta) \quad (3)$$

$$y = y_0 + v \Delta t \sin(\theta) \quad (4)$$

v is the vehicle speed (this approach considers a constant speed), l is the wheelbase, r is the turning radius, ω is the yaw rate, θ is the orientation, x, y are the positions in a Cartesian coordinate system, Δt is the time interval, and δ is the steering angle.

The candidates are generated based on constant rate change of the steering angle ($\frac{d\delta}{dt}$) and with a change to an opposite-sign value (its maximum value is denoted as $\frac{d\delta}{dt}_{max}$). For example based on Fig 5, 6 candidates are generated from $-\frac{d\delta}{dt}_{max}$ and $\frac{d\delta}{dt}_{max}$, 3 values are constant and 3 change from a positive or negative value to another one with opposite-sign (at fraction of the maximum evaluation time).

4) *Control Longitudinal Linear*: This function resolves the longitudinal vehicle control and it is also part of the control module. It generates a throttle of brake signal based on a function that relates the speed error with an action value. Nevertheless, it includes a fuzzy logic controller explained in previous articles [9]. In the future, this controller will be updated with a newer version based on a model.

B. Interfaces

For this article, the interfaces are all the environments where AUDRIC2 can be executed. The modules can be integrated with different environments or platforms because they are programmed in C++, and it has 2 function wrappers around ROS and MATLAB/Simulink. In the future, we are planning to integrate the modules in ROS2 as well.

1) *ROS*: This interface supports all the modules previously mention on section II-A. Nevertheless, it includes interfaces (bridges) to interact with the perception module of Autoware.ai [14]. Currently, it is supported the NDT matching algorithm used to generate the vehicle position based on LiDAR information, as well as the EKF filter. In terms of obstacle detection, the framework has a bridge to connect the obstacle detected with the Autoware clustering algorithm and its UKF algorithm to estimate its movement direction and speed. We are working to integrate a camera sensor that improves the detection of obstacles on the road.

2) *MATLAB/Simulink*: Simulink is a block diagram environment for multi-domain simulation and Model-Based Design. It supports system-level design, simulation, automatic code generation, and continuous test and verification of embedded systems. Thanks to the so-called MEX File Functions, all the modules developed in AUDRIC2 can be integrated since allows C/C++ programs to behave just like MATLAB functions. In Fig. 6 a more detailed implementation of AUDRIC2 in Simulink can be observed. Where each block corresponds to one of the six-module architecture described previously.

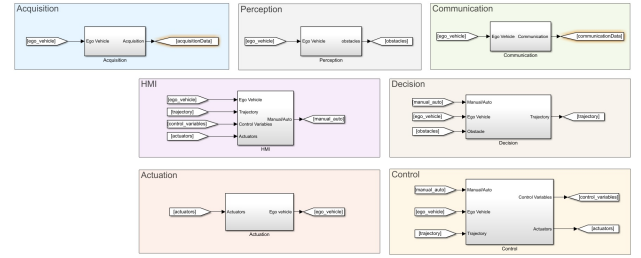


Fig. 6. AUDRIC2 Simulink block diagram.

Thanks to environmental compatibility, in addition to the developed functions, TCP/UDP bridges can be made to interact with outside sources e.j. external simulators. Therefore, developing a UDP bridge connection with the Dynacar [15] simulator can be carried out. By doing so, a reliable vehicle representation can be obtained while having a 3D interface for monitoring purposes [16].

Moreover to the aforementioned integration, connections with actuators or D-GPS can be made through CAN Toolbox already integrated with the MATLAB environment. Works using these features with the AUDRIC previous version can be observed in the following articles [17], [18], [19], where validation of merging, overtaking, and car following maneuvers are presented.

Taking into consideration, the progress made with the framework, developments in different maneuvers such as roundabout platoon merging are planned in the future. Moreover, with the integration of the Commsignia V2X devices, a new car following module version is under development, where vehicles with multiple dynamics (e.g. a 12m-long bus, Gulliver shuttle, or Renault Twizy 80) are being contemplated.

3) *Integration with CARLA simulator*: Released in 2017, CARLA simulator is one of the most important automated vehicle-oriented open-source simulators nowadays. The regular updates it receives from a highly active community give this software a great versatility when it comes to validating experimental algorithms [20], [21]. Its compatibility with ROS makes it easy to integrate with other platforms like Autoware [22], [23] or MATLAB [24]. Therefore, exists a big interest in testing AUDRIC2 in CARLA simulator. Our current developments are going towards this goal, which will be studied in detail in future works.

III. VALIDATION SCENARIO

The previous sections have highlighted the intention of using AUDRIC2 to control different vehicles (Fig. 1). The new vehicles that will use this solution will be a 12 meters electric bus and two electric shuttles into the framework of the European project SHOW.

In these terms, the current section contains the explanation of one of the sites proposed in the project SHOW and the first simulations conducted in this context.

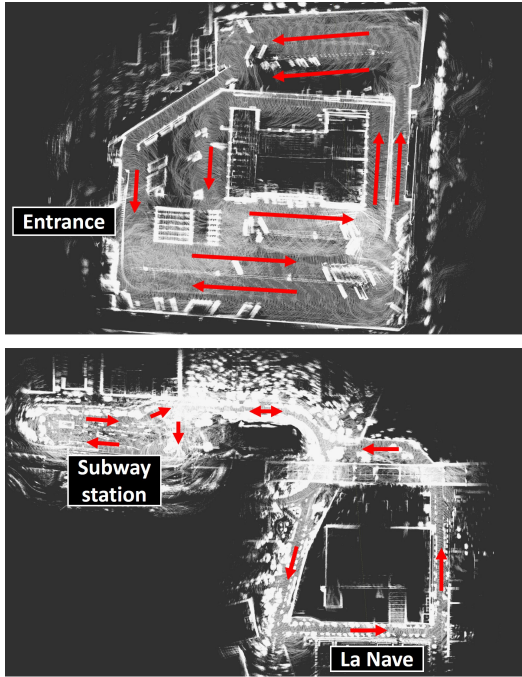


Fig. 7. Scenarios considered in the SHOW project, Madrid pilot

A. European project SHOW

SHOW is a European initiative which is intended to bring the automated operation of Personal Transport (PT) and Demand-Responsive Transport (DRT) services considering dense city traffic, peri-urban areas, and neighborhoods in 20 different European cities of 13 countries. The vehicles considered, in the project, are automated buses, robo-taxis, cars, cargo vehicles, and shuttles.

This article considers the Spanish pilot in the city of Madrid. It is separated into two different scenarios in the

same city, those are shown in Fig. 7 (3D Maps generated with Autoware which is used as a complement of the perception for the AUDRIC2 Framework). The upper part of the aforementioned figure is a depot of one of the companies in charge of public urban transport (Carabanchel depot), and the lower part is an open-traffic scenario that connects a subway station with the innovation hub of "La Nave" (Villaverde scenario), both in the Madrid region.

Platooning, teleoperation, automated docking/parking, and automated depot management are some of the use cases that will be tested in the Carabanchel depot. Automated passenger transport service, using shuttles, will be considered in the Villaverde scenario where a variety of difficult conditions will be present. Mixed traffic circumstances, regulated (traffic lights) and non-regulated intersections, narrow roads, high density of pedestrians, among others are some of the conditions present in this last scenario.

The vehicles used, in both scenarios, are shown in Fig. 1 and they are two automated Renault Twizy, one 12m electric bus, and two automated shuttle. Currently, the Twizy and the electric bus are receiving improvements in hardware and software, and the electric shuttles (Gulliver model) are in the process of setting up the actuators and the interface to control it automatically.

B. Preliminary tests: Basque Country

As it was mentioned in previous sections, the objective of this article is to present the updated version of the AUDRIC2 framework and its deployment in the SHOW project. Nevertheless, a real test will be considered in the final demonstration of the project (end of 2023). Moreover, the vehicles do not have the permits to drive in automated mode in Madrid.

In these terms, the first tests will be conducted at Tecnalia premises in the Basque Country (Spain) around the technological park, using the automated Renault Twizy platform. For this work, the tests are in simulation environments doing the definition of the scenario as close as possible to the real one.

The parameters used for the simulation, and the decision and control algorithm, are the following:

Veh. length	2.4 meters
Veh. width	1.4 meters
Veh. wheelbase	1.75 meters
Max. steering angle	32°
Max. time for turning	2 second

The system used to execute the algorithm is a Core i7-6820HQ with 32GB of RAM along with the ROS interface (ROS Melodic, Ubuntu 18.04 LTS).

Fig. 8, at its top, presents the test case conditions, around Tecnalia premises, that will be tested as a verification phase before facing Madrid scenarios. This scenario has an approximated length of 1.5km which is similar to the longest scenario of Madrid (Villaverde). It has a roundabout and a 90° left turn. The rest of the scenario is shaped by soft and

tight bends. Moreover, a variety of merging conditions will be considered and tested in future contributions.

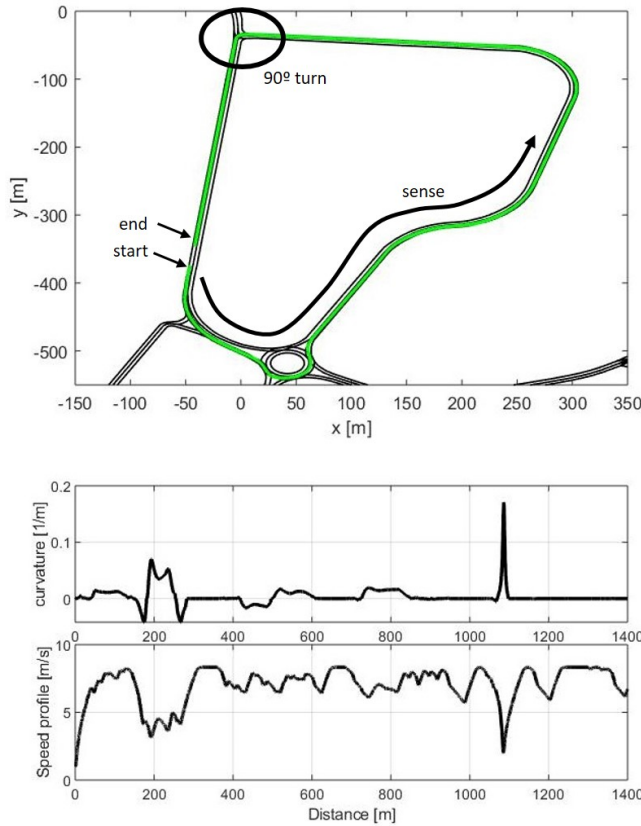


Fig. 8. First tests without obstacles.

The middle and lower part of Fig. 8 presents the curvature and speed profile respectively. Those values are extracted from the green trajectory presented at the top of the aforementioned figure. One of the main advantages of the *trajectoryGenerator* module is the smoothness of the curvature which improves the tracking of the steering control, as well as the generation of smooth and comfortable speed profiles.

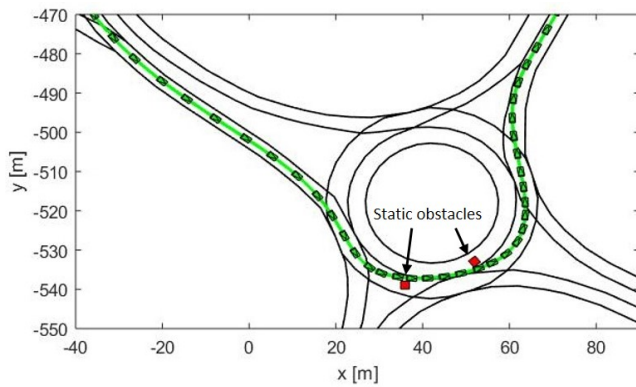


Fig. 9. Static obstacles in the roundabout.

Furthermore, Fig. 9 depicts the capacity of the framework to face obstacles under difficult circumstances such as on a bent segment or in the middle of a roundabout. It is

important to highlight that this work does not present moving obstacles due to a deeper study will be presented on an article dedicated to the *trajectoryGenerator* module. For further information about moving obstacles, a method similar to the one previously presented in [18] is used in this new version of the algorithm.

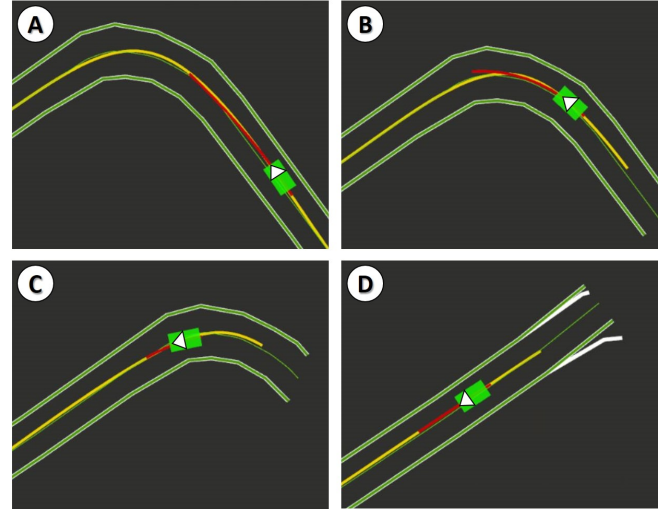


Fig. 10. View of the trajectory generation and control on the left turn.

Fig. 10 depicts the capacity of manage a 90° left turn. The original map is depicted in white, the boundaries of the drivable space are green, the generated trajectory is in yellow and the time propagation of the lateral control (steering wheel) is in red. The length of the control prediction reduces as the vehicle arrives at the intersection, and it increases when the vehicle exits the turn. This is mainly produced by the speed profile which is associated with the curvature of the trajectory (refer to Fig. 8 between a distance of 1000 and 1200 meters).

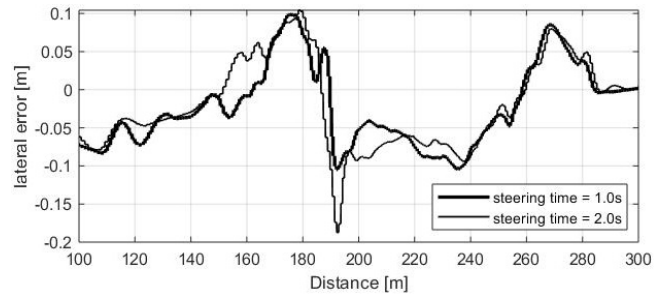


Fig. 11. Analysis of the steering delay and its effect on real test cases.

The incidence of the turning time (time to move the steering wheel from straight position to the maximum at the right or left side which is a value used to calculate the $d\delta/dt_{max}$ used by the *Control Lateral Model Based*) in the response of the controller is appreciated in Fig. 11. The time of moving the steering wheel has been simulated to generate a close-to-reality result. Mostly, the lateral error between the vehicle position and its desired trajectory is under 10cm in

most of the cases. Nevertheless, this error increases along with the turning time of the steering wheel.

In these terms, the control values will be studied in more detail in the real vehicle due to simulated values do not provide enough information to obtain a conclusion about the performance of the control method. However, it is important to highlight that the expected result is close to this one because the steering wheel rate has been considered in the controller.

IV. CONCLUSIONS AND FUTURE WORKS

This article has presented an updated version of an automated driving framework that was proposed in 2017. It has included a novel interface to ROS along with the previous one (based only on MATLAB/Simulink). Moreover, this new version of the AUDRIC2 framework will be embedded in different vehicles, such as cars, shuttles, buses, and non-holonomic mobile robots. This is an important contribution because it can prove the capacity to be a scalable framework considering vehicles with different dynamics and geometries.

The article highlights that AUDRIC2 is aiming at decision-making and control algorithms for ground automated vehicles which demands the integration of a 3rd party software to obtain vehicle and obstacle information (mainly in the perception stage). In these terms, the current version of AUDRIC2 can be combined with the Autoware framework, specifically Autoware.AI (based on ROS). This integration will be explained in an upcoming article along with real vehicle tests.

In terms of future work, real-platform tests have been programmed for the upcoming month as well as the creation of the ROS2 interface that permits the integration of the newest version of Autoware.Auto. Moreover, the weather conditions of the Basque Country will be exploited to make more robust and resilient algorithms under foggy or rainy conditions.

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